Native Binary-Weight Training Outperforms Zero-Shot Sign Projection in MNIST Diffusion

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***Abstract*—Does severe failure after projecting a diffusion model to one-bit weights show that its binary representation is unusable? We test that implication, rather than propose a new quantizer, in a small MNIST residual U-Net. We compare floating-point training, zero-shot per-channel sign/scale projection of a trained model, binary-weight training from random initialization, floating-point warm-started quantization-aware training (QAT), and a binary-weight/ binary-activation core. The primary FP32, native W1A32, and projected W1A32 comparison fixes the parameterization, data, training budget where applicable, and 1,000-step ancestral sampler, with three independent training seeds. Native W1A32 training obtains domain-feature Fréchet distance 46.33 ± 17.24, 28.4-fold lower than matched zero-shot projection (1314.20 ± 314.96), but remains substantially worse than FP32 (3.15 ± 0.38). Warm QAT also recovers much of the projection loss, while the W1A1 core retains a large distributional gap. An elementary feasible-set argument explains why one projected floating-point solution is not an impossibility certificate for the binary hypothesis class. The supported claim is deliberately local: on MNIST, optimizing binary forwards reaches a much better W1A32 solution than this zero-shot projection baseline; this does not establish full W1A1 success, floating-point parity, or transfer to natural images.**

***Index Terms*—binary neural networks, diffusion models, post-training quantization, quantization-aware training, MNIST**

I. Introduction

Diffusion models repeatedly evaluate a time-conditioned denoiser, making model memory and arithmetic attractive compression targets. Binary neural networks replace many real-valued weights by signs and optional scales [1], [2]. At the extreme, BiDM studies binary weights and activations (W1A1) and motivates specialized architecture and distillation by the severe degradation of naive full binarization [3]. BinaryDM studies accurate weight binarization at W1A4 [4].

A narrower logical question precedes those system designs: if a trained FP32 denoiser collapses after sign projection, has the binary-weight representation failed, or has a point optimized for a different forward model simply been projected poorly? Our zero-shot post-training baseline evaluates only the projected FP32 solution; it is not a calibrated PTQ method. BinaryConnect-style training instead maintains latent real parameters while using binary weights in every forward pass [5]. The two procedures need not reach the same binary point.

We isolate this distinction on MNIST. The primary precision conditions share parameter names, shapes, U-Net graph, sampling procedure, and evaluation; all trained stages use the same optimizer and 12-epoch budget. The experiment supports three findings:

- native W1A32 training reduces dFID by 28.4-fold relative to matched zero-shot projection, so projection failure is not a certificate of binary-weight representational failure in this setting;
- warm-started QAT also recovers substantially, showing that random initialization is not uniquely necessary once binary forwards are optimized; and
- FP32 remains much better, while a W1A1 core remains weak; the result therefore neither contradicts the difficulty identified by BiDM nor establishes a general binary-diffusion solution.

The contribution is a controlled empirical distinction between projection and optimization. The sign/scale quantizer, straight-through estimator (STE), and residual components are standard and are not claimed as algorithmic novelty.

II. Related Work

Diffusion PTQ requires more care than applying an image-classification quantizer unchanged. Q-Diffusion uses timestep-aware calibration and separate treatment of shortcut activations [6]; PTQD models and corrects quantization noise accumulated through denoising [7]. These methods target multi-bit PTQ and are substantially more sophisticated than our deliberately uncalibrated sign-projection baseline. BinaryDM combines an evolvable binary basis with representation mimicking for W1A4 diffusion [4], whereas BiDM combines binary-specific structure and distillation for W1A1 diffusion [3]. We do not claim competitive image quality or efficiency against these systems.

Binary-network research likewise shows that training procedure and activation distributions matter. BinaryConnect optimizes latent real weights under binary forwards [5]; XNOR-Net introduces channel/filter scaling [2]; Real-to-Binary Net uses a staged training schedule [8]; and ReActNet learns activation shifts and reshaping [9]. Our experiment uses only the standard sign/scale and STE ingredients so that the intervention remains the training path rather than a new binarizer.

III. Projection and Constrained Optimization

Let Θ_F be a floating-point parameter space, Θ_B the feasible binary-weight space, $\mathcal{L}$ the diffusion training objective, and $P : \Theta_F \to \Theta_B$ the baseline sign/scale projection. Let $\theta_F^\star$ minimize $\mathcal{L}$ over Θ_F and $\theta_B^\star$ minimize it over Θ_B.

a) Feasible-set observation: Assuming the minima exist, the loss of a projected floating-point optimum is an upper bound on the optimal binary-constrained loss:

$$\mathcal{L}(\theta_B^\star) \le \mathcal{L}(P(\theta_F^\star)). \tag{1}$$

Because $P(\theta_F^\star) \in \Theta_B$, it is one feasible point in the minimization defining $\theta_B^\star$. Consequently, poor performance of $P(\theta_F^\star)$ alone cannot establish that every member of Θ_B performs poorly. This is a direct consequence of the definition of a minimum, not a new optimization theorem.

This elementary statement is not a convergence result. Native STE training is not guaranteed to find $\theta_B^\star$, and diffusion quality is not determined by training loss alone. It identifies exactly what the experiment must test: whether optimization under binary forwards reaches a materially better feasible point than zero-shot projection.

IV. Controlled MNIST Experiment

A. Denoiser and binary forward

For clean image x_0, noise ϵ, and timestep t, every condition minimizes the DDPM noise-prediction objective [10]:

$$\mathbb{E}_{x_0,t,\epsilon} \left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t \right) \right\|_2^2 . \tag{2}$$

The preactivation denoiser in Fig. 1 is a residual U-Net [11] with base width 16, widths $[16, 32, 64]$, two downsampling stages, two skip concatenations, GroupNorm, and sinusoidal time embeddings [12], [13]. It has 218,993 trainable parameters.

For output filter W_o, the primary uncentered binary projection is

$$\widehat{W}_o = \alpha_o \operatorname{sign}(W_o), \qquad \alpha_o = \operatorname{mean}(|W_o|). \tag{3}$$

Latent FP32 parameters receive identity-STE gradients [14]. The centered ablation subtracts the filter mean before sign and scale. The W1A1 core additionally signs block activations with a clipped-identity backward pass. Input/output convolutions, normalization, time conditioning, biases, scales, accumulation, and residual additions remain FP32; "W1A1 core" therefore does not mean an end-to-end binary appliance.

B. Conditions and optimization

FP32 and native conditions train from random initialization for 12 epochs. The zero-shot projection baseline loads the corresponding completed FP32 checkpoint, changes only the convolutional forward weights, and performs neither calibration nor optimization. Warm QAT loads that checkpoint and adds a full 12-epoch binary-forward stage, for 24 total epochs. Native W1A32 crosses centered/uncentered projection with pre-/post-activation residual ordering [15]; the headline matched comparison is uncentered and preactivation. Moving normalization for the postactivation ablation changes its GroupNorm affine tensors, yielding 218,897 parameters rather than 218,993; it is therefore an architectural ablation, not part of the exactly parameter-compatible precision comparison.

We use the canonical 60,000/10,000 MNIST split [16], scale images to $[-1, 1]$, and train with AdamW [17], learning rate 2×10^{-4}, weight decay 10^{-4}, cosine decay, gradient clipping at 1, batch size 256, and training seeds $\{0, 1, 2\}$. Sampling uses all 1,000 ancestral DDPM steps. Each reported distribution contains 2,000 samples from a fixed sampling seed independent of the training seed.

C. Evaluation and controls

An independently initialized MNIST classifier, trained only on the real training split, reaches 98.03% test accuracy and 1.44% ten-bin expected calibration error [18]. Its penultimate features define domain-feature Fréchet distance (dFID) [19], unbiased polynomial-kernel distance (dKID) [20], and $k = 5$ density and coverage [21]. We also report Jensen–Shannon divergence of predicted class priors [22]. These domain-specific values are not comparable to ImageNet FID. Finite-sample Fréchet estimates are biased [23]; with only 2,000 generated samples per seed, our dFID values are descriptive effect-size summaries rather than population estimates or significance tests.

A disjoint real–real split gives dFID 0.076, density 0.969, and coverage 0.971. Clamped Gaussian noise gives dFID 2278.24 and zero density/coverage. Although that noise receives mean peak classifier confidence 1.000, confidence is not used as a quality metric.

V. Results

a) Native optimization outperforms zero-shot projection: Table I gives the main comparison. Native W1A32 has 28.4-fold lower mean dFID than its matched projection configuration and nonzero density and coverage. This is an empirical witness that a binary-forward optimization path can reach a substantially better model than sign-projecting the FP32 endpoint. It does not show that all binary training succeeds: the standard deviation is large and every reported distribution metric remains worse than FP32.

b) Warm QAT also recovers: Warm QAT obtains dFID 66.45 ± 28.49, 13.1-fold below the mean of its matched centered projection baseline (871.12 ± 717.34). The projected baseline is highly variable, and warm QAT uses an additional 12-epoch stage, so this ratio is descriptive rather than a sample-efficiency claim. Its centered configuration is not a strictly matched comparison to the best uncentered native row. Native training is superior to zero-shot projection, not necessarily to every warm-start or teacher-dependent procedure.

c) W1A1 remains difficult: The W1A1 core improves over both zero-shot W1A32 projections in dFID but has density 0.005 and coverage 0.009. Our experiment therefore does not refute BiDM's observation that simultaneous weight and activation binarization is difficult. It instead isolates a weaker but useful statement about one-bit weights with FP32 activations.

d) Architecture does not yield a universal rule: Table II shows an interaction: preactivation helps only without centering. Centering also raises mean latent-to-projected angular error from 33.29° to 34.05° in that pair. Neither choice is therefore a universal safeguard against degradation.

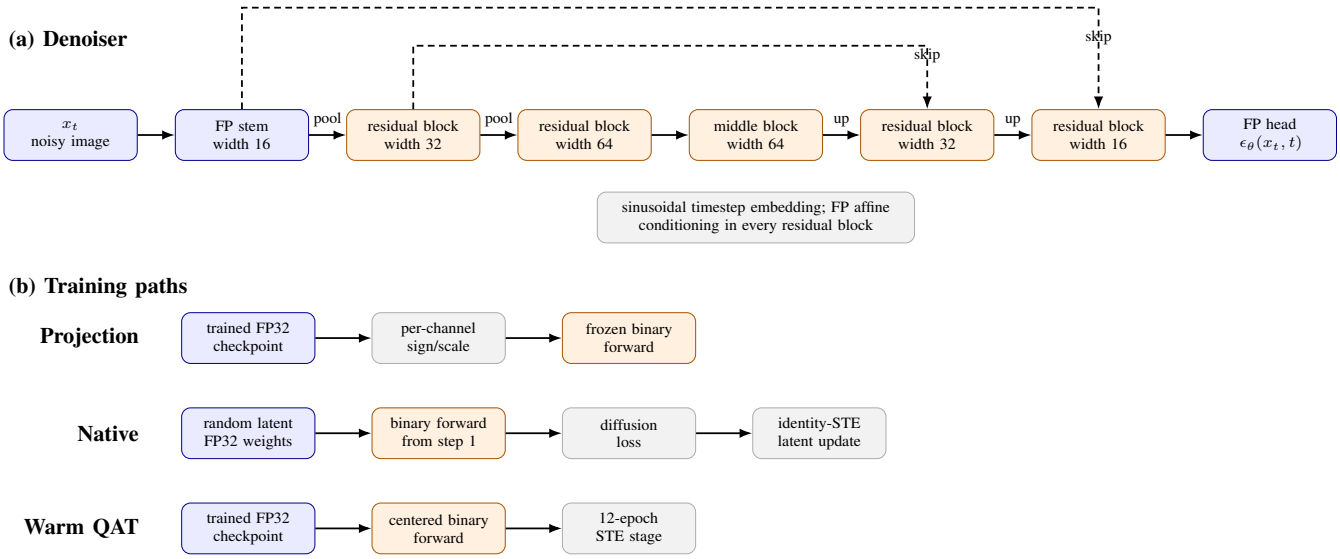
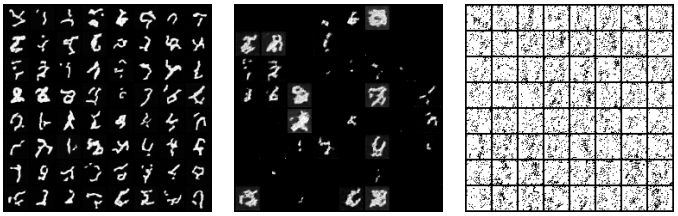


Fig. 1. Implementation-derived diagram. (a) Preactivation U-Net used by the primary comparison; dashed routes are the actual stem-to-final-decoder and 32-channel encoder-to-first-decoder concatenative skips. Orange blocks use W1A32 unless the W1A1 core is selected; normalization, conditioning, scales, residual additions, and boundaries remain FP32. (b) Compared optimization paths. The projection baseline performs no calibration or parameter updates.

TABLE I

MNIST METRICS, MEAN ± SAMPLE STANDARD DEVIATION OVER THREE TRAINING SEEDS. EACH SEED CONTRIBUTES 2,000 SAMPLES FROM THE SAME ANCESTRAL SAMPLER. BOLD MARKS THE BEST NON-FP CONDITION, NOT PARITY WITH FP32.

Condition	dFID ↓	dKID ↓	Density ↑	Coverage ↑	Class JS ↓
FP32	3.15 ± 0.38	0.056 ± 0.013	0.284 ± 0.014	0.195 ± 0.017	0.0329 ± 0.0091
Native W1A32, uncentered/pre	$\mathbf{46.33 \pm 17.24}$	$\mathbf{1.286 \pm 0.816}$	$\mathbf{0.022 \pm 0.005}$	$\mathbf{0.033 \pm 0.010}$	$\mathbf{0.2946 \pm 0.0202}$
Warm QAT W1A32, centered/pre	66.45 ± 28.49	4.945 ± 3.686	0.019 ± 0.004	0.032 ± 0.004	0.3196 ± 0.0593
Native W1A1 core	204.55 ± 69.77	41.03 ± 25.01	0.005 ± 0.002	0.009 ± 0.004	0.3395 ± 0.0268
Projected W1A32, centered/pre	871.12 ± 717.34	1581.31 ± 2433.16	0	0	0.5326 ± 0.0002
Projected W1A32, unc./pre	1314.20 ± 314.96	2467.50 ± 1549.51	0	0	0.5314 ± 0.0023



(a) FP32 (b) Native W1A32 (c) Projected W1A32

Fig. 2. Training seed 0 and sampling seed 4242. Native binary-weight training retains digit-like structure but remains visibly below FP32; zero-shot sign projection produces non-digit structure.

TABLE II

NATIVE W1A32 ARCHITECTURAL ABLATION (MEAN dFID; LOWER IS BETTER).

Residual order	Uncentered	Centered
Pre-activation	46.33	99.50
Post-activation	84.26	82.46

VI. SCOPE, LIMITATIONS, AND REPRODUCIBILITY

This is an MNIST existence result for a small unconditional model. Three seeds expose high variance but are insufficient for fine-grained significance claims; 2,000 samples per seed make absolute metric estimates noisier than common 50,000-sample evaluations. A separate compact CIFAR-10 attempt is intentionally excluded from the evidence because its FP32 control also failed, so it could not adjudicate binary versus floating-point quality on natural images. No theorem here establishes STE convergence, score approximation, or scaling behavior.

Binary convolutions contain 195,328 of 218,993 parameters. Counting one bit per binary weight plus FP32 per-output-channel scales and all remaining FP32 tensors gives a calculated raw payload of 121,316 bytes versus 875,972 bytes for FP32, a $7.22\times$ reduction. This excludes container and runtime overhead and is not a hardware measurement: ordinary framework convolutions still execute floating-point operations, and we claim no latency or energy improvement.

The released pipeline [24] records configurations, seeds, environment versions, the exact experiment commit, checkpoint SHA-256 values, histories, metrics, controls, and a packed export. The repository README maps the exact experiment

entry points and MNIST artifact directories. Tests cover parameter compatibility, forward shapes, diffusion schedules, metric controls, and physical packing.

VII. Conclusion

Zero-shot sign projection answers whether one particular FP32 endpoint survives binarization; it does not answer whether binary-forward optimization can find a better binary-weight solution. In a controlled MNIST diffusion model, native W1A32 training reduces domain-feature distance by 28.4-fold relative to matched projection, while warm QAT also recovers substantially. The result is meaningful but bounded: FP32 remains much better, W1A1 remains weak, and natural-image transfer is unresolved. Future binary-diffusion comparisons should distinguish native, warm-started, distilled, and zero-shot training paths rather than treating zero-shot projection failure as the limit of the representation.

Code availability. Source, checkpoints, packed weights, and evaluation artifacts are available at https://github.com/elixpo/monodiffuse.

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